

Ryan Cho

Senior Software Engineer — Identity, Authentication & Platform Systems

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PROFESSIONAL SUMMARY

Software engineer with 8+ years delivering secure authentication, identity platforms, and API services across finance and SaaS. Architected centralized login flows, OAuth/OIDC integrations, and headless identity modules consumed by mobile and web clients. Led CIAM onboarding, token lifecycle controls, and security hardening for regulated production systems. Known for full-stack ownership, SDK-friendly service design, and cross-functional technical leadership.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Owned a **centralized authentication platform** that unified login, session management, and token refresh for web and mobile-facing services, reducing duplicate identity logic across 12 product surfaces.
- Designed **headless identity APIs and SDK-ready modules** enabling partner and internal teams to embed secure sign-in without rebuilding auth flows, cutting integration time for new client applications by roughly 40%.
- Led **CIAM onboarding** for enterprise identity providers, mapping user lifecycle events, consent policies, and federation rules into production services used by portfolio and investor-facing applications.
- Hardened **mobile security controls** including certificate pinning guidance, secure token storage patterns, and refresh rotation policies adopted by client engineering teams consuming platform authentication services.
- Built **OAuth 2.0 and OIDC authorization services** with scoped access tokens, role-based claims, and audit logging, supporting thousands of daily authenticated sessions across regulated financial workflows.
- Partnered with security and compliance stakeholders to implement **threat modeling, penetration test remediation, and MFA rollout**, closing critical findings ahead of external audit deadlines.
- Mentored 4 engineers on **identity architecture, API contract design, and secure coding practices**, improving review quality and reducing authentication-related production incidents over two release cycles.
- Drove cross-team adoption of a **shared login gateway** through design reviews and migration playbooks, deprecating legacy session cookies and inconsistent credential handling across legacy services.

Technology Analyst, Incoming Technology Analyst, Technology Summer Analyst,

May/2019 - Apr/2022

BNP Paribas

Remote | Seventh Avenue, New York

- Delivered **enterprise authentication enhancements** for internal banking applications, consolidating LDAP and federated login paths into a standardized access layer used by trading and operations teams.
- Implemented **API security middleware** with JWT validation, service-to-service authorization, and request signing, protecting high-volume transaction endpoints processing millions of requests weekly.
- Supported **mobile and web client integration** by publishing versioned identity contracts and integration guides, enabling downstream teams to adopt centralized login without custom credential handling.
- Automated **CI/CD security gates** including dependency scanning, secret detection, and static analysis, reducing vulnerable builds reaching staging environments by a measurable margin each quarter.
- Built monitoring dashboards for **authentication failures, lockouts, and anomalous access patterns**, giving operations teams faster detection of credential abuse and misconfigured client integrations.
- Collaborated with infrastructure teams to deploy **identity services on AWS** with encrypted secrets management, high availability failover, and disaster recovery runbooks aligned to bank policy.
- Translated regulatory and InfoSec requirements into **technical controls for session timeout, password policy, and privileged access**, documenting evidence for internal compliance reviews.
- Coordinated **cross-functional delivery** across risk, product, and engineering groups during major platform upgrades, maintaining service continuity while migrating users to updated login experiences.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Built **user authentication flows** with email verification, password reset, and session persistence for a customer-facing web product serving early beta users.
- Integrated **third-party OAuth providers** to streamline account creation while preserving secure token exchange and profile synchronization across application layers.
- Developed **REST APIs and React UI components** for account settings and permission-aware navigation, improving consistency between frontend state and backend authorization checks.
- Added **automated test coverage** for login edge cases, expired sessions, and invalid credentials, catching regressions before weekly releases.
- Documented **API authentication patterns** and local setup steps for new contributors, shortening onboarding time for additional intern and contractor developers.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Implemented **secure member login and role-based page access** for a nonprofit web portal, ensuring authenticated users reached only authorized content and administrative tools.
- Built responsive **HTML, CSS, and JavaScript interfaces** connected to backend credential validation, delivering a stable sign-in experience across desktop and mobile browsers.
- Configured **HTTPS, secure cookies, and server-side session handling** to protect user credentials and reduce exposure to common web authentication vulnerabilities.
- Resolved **production login defects and performance bottlenecks** during launch support, restoring reliable access for staff and community users under tight delivery timelines.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, JavaScript, HTML/CSS, Mobile Web Integration, OAuth Client Flows

Backend: Java, Spring Boot, Node.js, Python, REST/gRPC APIs, Identity Services, OAuth 2.0/OIDC

Databases: PostgreSQL, MySQL, Redis, MongoDB

Cloud & DevOps: AWS, Docker, Kubernetes, Jenkins, GitHub Actions, Terraform, Secrets Management

Monitoring & Tools: Datadog, Splunk, Prometheus, Postman, SonarQube, Auth Debugging & Token Analysis

Workflow & PM: Agile/Scrum, Jira, Confluence, Cross-Functional Planning, Technical Documentation, Stakeholder Alignment